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Contacts:

CIFREL

Università Cattolica del Sacro Cuore

Via Necchi 5

20123 Milano

Telephone: 0039.02.7234.2976

e-mail: cifrel@unicatt.it

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The Promised Land? Municipal Electoral Cycles and Land Consumption in Italy.

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Giovanni Cuttica¹, Leonzio Rizzo^{1,2}, Luisa Loiacono^{1,3}, and Riccardo Secomandi^{1,2}

¹Department of Economics, University of Ferrara

²Institut d’Economia de Barcelona (IEB), Spain

³London School of Economics Visiting Fellow, London, United Kingdom

Abstract

Does the electoral calendar shape construction activity in Italian municipalities? We study this using the ISPRA *Carta Nazionale del Consumo di Suolo* — the first satellite-derived land-cover dataset applied to a political business cycle study — covering 7,761 municipalities over 2015–2024. Unlike fiscal and administrative outcomes, ISPRA measurements cannot be retroactively altered by the governments under study. To handle the recurring nature of electoral treatment, we implement a Stacked DiD design with cohort-specific clean control groups and a prior-cycle contamination guard. We find a *post-election mandate cycle* in construction sites: mayors suppress construction in the year before the election and accelerate it in the two years that follow, a swing of approximately 12 percentage points. The pattern reverses among lame-duck mayors, consistent with a political-debt mechanism, and is concentrated in municipalities with fewer than 3,000 inhabitants and in northern Italy.

JEL Classification: D72, R52, H72

Keywords: Political Business Cycles, Land Use, Italian Municipalities, Stacked Difference-in-Differences, Electoral Cycles

1 Introduction

Does the electoral calendar shape construction activity in Italian municipalities? Land use — the conversion of agricultural, natural, or green space into built-up or sealed surfaces — is an attractive electoral instrument: it is administratively controlled at the municipal level, generates visible local activity (construction cranes, excavations, newly paved roads), and can in principle be timed to coincide with the electoral calendar. Yet credible causal evidence on whether such cycles exist in land-use policy is absent from the literature, held back by two problems: the use of outcome variables that are themselves administratively reported by the governments under study, and an identification strategy that does not account for the recurring nature of electoral treatment. This paper addresses both.

Setting and motivation. Italian municipalities hold mayoral elections every five years on a staggered calendar, with each municipality’s election date determined by the preceding one. This generates within-country variation in electoral timing while holding institutions constant. Since the 2008 abolition of the primary-residence property tax (ICI), fiscal manipulation space for Italian mayors has been progressively compressed by central government rules (Ferraresi et al., 2019; Giacobbe et al., 2024): before the reform, municipalities in election years spent roughly 19 euro per capita more than non-election-year municipalities; after the reform this cycle narrowed markedly. The compression creates a structural incentive to redirect electoral activity toward regulatory channels that are not subject to fiscal rules. Land-use authorisation is precisely such a channel: mayors retain direct authority over building permits (*permesso di costruire*) and road infrastructure licences, and the timeline between a permit decision and physically observable site activity is short enough to generate within-term variation detectable in annual satellite data.

Data contribution. We use the ISPRA *Carta Nazionale del Consumo di Suolo* (CNCS), a nationally consistent raster product derived from multi-temporal satellite imagery (primarily Sentinel-2 and SPOT) at 10 m spatial resolution, updated annually from 2015 onwards. To the best of our knowledge, this is the first application of CNCS data to a political business cycle study. The administrative independence of the source is methodologically important. All existing PBC papers — including the closest empirical predecessor, Ferraresi et al. (2024), who study road investment in the same Italian staggered-election setting — use fiscal or administrative outcome variables that are defined and reported by the same governments whose behaviour is under study. If incumbents manipulate building permits as an electoral instrument, permit-based counts are endogenous to the very behaviour we wish to measure.

ISPRA land-cover classifications are derived from satellite observation: an incumbent can delay a permit, but no mayor can edit a Sentinel-2 image.

Identification contribution. The standard approach in the PBC literature is a two-way fixed effects (TWFE) event study. TWFE has two problems in the Italian municipal setting. First, with staggered election timing, municipalities that have already voted serve as implicit controls for those voting later, generating biased estimates when treatment effects are heterogeneous across cohorts (Baker et al., 2022; Goodman-Bacon, 2021). Second, and more fundamentally, Italian elections are a *recurring pulse*: every municipality is treated every five years, and treatment status returns to zero after each cycle. This violates the irreversibility condition that underpins the Callaway–Sant’Anna (Callaway and Sant’Anna, 2021) and Sun–Abraham (Sun and Abraham, 2021) estimators — both require a never-treated comparison group that does not exist when every municipality eventually faces an election. We implement a Stacked DiD design (Baker et al., 2022; Wing et al., 2024) with cohort-specific clean control groups and a prior-cycle contamination guard specific to recurring electoral cycles. A systematic review of 13 recent PBC papers (2020–2025) confirms that no existing study has applied such a correction to an electoral-cycle setting.

Main findings. We find a *post-election mandate cycle* concentrated in construction sites. Construction activity is suppressed relative to the election year in the year before the election ($\hat{\delta}_{-1} = -0.050^{**}$, conditional specification), then surges in the two years that follow ($\hat{\delta}_{+1} = 0.042^*$, $\hat{\delta}_{+2} = 0.068^{**}$). Re-normalised to the pre-electoral trough, the cycle amplitude is approximately 12 percentage points. As expected, no cycle appears for longer-run stock categories (total consumption, buildings, paved roads, other sealed surfaces, permanent consumption) whose measurements reflect completions rather than current permit decisions. The cycle is driven by first-term mayors facing a binding re-election constraint: lame-duck mayors (second consecutive term, cannot run again) exhibit the *opposite* pattern, increasing construction before the election rather than suppressing it. This sign reversal at the term-limit boundary is the sharpest identification result in the paper: it rules out any mechanical or seasonal interpretation of the pre-electoral lull. The cycle is also entirely concentrated in municipalities with fewer than 3,000 inhabitants — where mayors retain direct administrative control over the permitting process — and in northern Italy.

Roadmap. Section 2 reviews the PBC and land-use literatures. Section 3 describes the Italian institutional setting. Section 4 presents the data. Section 5 lays out the identification strategy. Section 6 reports results and discusses the findings. Section 7

concludes.

2 Literature Review

2.1 Political Business Cycles: Theory and Empirics

The modern PBC literature traces its origin to Nordhaus (1975), who modelled incumbents as strategically manipulating macroeconomic outcomes (inflation, unemployment) to maximise votes. The subsequent 40 years of research are surveyed in Dubois (2016), who identifies a fundamental distinction between *outcome-based* cycles (manipulation of GDP, inflation, or unemployment, largely absent in developed democracies) and *instrument-based* cycles (manipulation of fiscal or regulatory instruments). The theoretical underpinning of the whole approach, treating elected officials as utility-maximising agents responding to electoral incentives, is rooted in the public choice tradition (Ostrom and Ostrom, 1971).

The sub-national and municipal level has emerged as the most fruitful testing ground for PBC hypotheses, in part because local governments have direct control over visible investment categories. Veiga and Veiga (2007) study all mainland Portuguese municipalities over 1979–2001 and document pre-electoral increases in capital expenditure. Their setting features nationally synchronised elections, unlike the staggered Italian cycles examined in this paper, but the mechanism is identical: capital spending is more susceptible to electoral manipulation than current expenditure. At the European level, Efthyvoulou (2011) applies the analysis to national governments and confirms pre-electoral fiscal deficits, with the effect attenuated in proportional-representation systems and further compressed in mature democracies. For Italy specifically, the literature has been shaped by a series of papers exploiting two key institutional features: the 2008 abolition of the primary-residence property tax, and the 5,000-inhabitant threshold of the Domestic Stability Pact.

Ferraresi et al. (2019) show that the 2008 ICI abolition substantially attenuated pre-electoral spending cycles: before the reform, municipalities in election years spent approximately 19 euro per capita more than non-election-year municipalities, financed mainly through user fees; after the reform, this cycle narrowed markedly. This finding is directly relevant to the present paper: the entire observation window (2015–2024) falls in the post-reform period, when fiscal manipulation space had already been compressed. This compression creates a structural incentive for incumbents to redirect electoral manipulation to non-fiscal, regulatory channels such as land-use permitting decisions. Bracco et al. (2024) confirm the same mechanism and extend it to the composition of expenditure. They find that Italian road investment rises 26% at $t - 1$ and 40% in municipalities with high construction-firm density. Road

investment, however, does not reduce accident rates, confirming that the increase is politically motivated rather than welfare-maximising. [Ferraresi et al. \(2024\)](#) further document that road investment rises monotonically throughout the term (26% at $t - 1$, 21% at $t - 2$, 19% at $t - 3$), with local procurement probability rising 52 percentage points in high-construction-density municipalities. This paper is the closest existing work to the present study: it uses the same institutional setting, and a physical infrastructure outcome adjacent to the land-use outcomes examined here.

Two further papers motivate the focus on fiscal constraints as a driver of non-fiscal substitution. [Giacobbe et al. \(2024\)](#) use a difference-in-discontinuities design at the 5,000-inhabitant DSP threshold and find that capital expenditure cycles are 136% larger for municipalities not subject to the Pact, those not constrained by fiscal rules. [Kluza et al. \(2024\)](#) document for Polish municipalities that off-budget debt rises at $t - 1$ when on-budget expenditure is constrained. Both findings are consistent with the hypothesis that land-use decisions (a regulatory channel largely outside the DSP perimeter) serve as a substitute electoral instrument when fiscal manipulation is constrained.

2.2 Land Use, Soil Sealing, and Political Incentives

The literature directly linking electoral cycles to land-use outcomes is thin but suggestive. A notable feature of the entire PBC empirical literature — including the Italian papers reviewed above — is that outcome variables are invariably *fiscal or administrative*: tax rates, expenditure categories, public procurement, off-budget debt. These variables are defined and reported by the same governments whose behaviour is under study, creating an obvious measurement concern. This paper takes a different approach: we use physical land-use data derived from multi-temporal satellite imagery, which is produced independently of any municipal administration. An incumbent can delay a building permit, but cannot edit an image taken from a satellites.

[Bimonte and Stabile \(2019\)](#) study Italian regions and establish that soil sealing follows an inverted-U Environmental Kuznets Curve with per-capita income, and that corruption increases sealing at low income levels. This is the only paper directly studying soil sealing in Italy; it establishes that land consumption responds to political and institutional incentives at the Italian regional level, and provides the empirical basis for including income controls in municipality-level regressions.

At the national level, [Kapur and Vaishnav \(2011\)](#) document that construction activity in Indian states contracts in election months, as builders redirect funds to political campaigns. This establishes the reciprocal mechanism: builders fund campaigns, and incumbents reciprocate with favourable land-use decisions after

the election. [Kurosaki et al. \(2024\)](#) provide a direct analogue: Indonesian village incumbents increase the sharecropping share of land contracts closer to the next election, using land decisions as clientelistic instruments.

The institutional context of Italian land governance amplifies the scope for such manipulation. Municipal governments retain primary authority over zoning plans (*Piano Regolatore Generale*) and building permits. Approval processes for construction sites and road-paving projects are administratively flexible, and the timeline between political approval and physical activity is sufficiently short to generate within-term variation detectable in annual land-use data.

3 Institutional Background

Italian municipalities (*comuni*) are governed by a directly elected mayor (*sindaco*) and a municipal council (*consiglio comunale*). The electoral term lasts five years. Municipal governments in Italy retain primary authority over spatial planning. The *Piano Regolatore Generale* (PRG) designates land use at the municipal level and is approved by the municipal council. Individual building permits (*permesso di costruire*) and excavation licences for infrastructure works are issued by municipal technical offices at the mayor’s discretion.

A mayor may serve at most two consecutive terms: incumbents in their first term have a binding re-election incentive, while second-term mayors are term-limited. This is a key heterogeneity dimension, and we exploit in this paper whether the mayor faces a binding re-election constraint. Under Italian municipal law, a mayor serving a second consecutive term is barred from standing again: the re-election incentive is present only for first-term incumbents. [Bordignon et al. \(2017\)](#) exploit exactly this variation in a DiD design comparing first- and second-term mayors before and after the 1999 PIT surcharge reform, finding that eligible (first-term) mayors raised total revenues 8% more than ineligible (second-term) mayors after the reform. [Coviello and Gagliarducci \(2017\)](#) document a complementary mechanism: longer mayoral tenure reduces bidder competition in public procurement and raises the probability of repeated contractor selection by 25 percentage points. Second-term mayors, freed from re-election concerns but with stronger contractor ties, may exhibit different land-use patterns — either a legacy-spending increase (rewards to aligned contractors) or an actual decrease (no electoral return on permitting activity).

Elections are staggered across municipalities: each municipality holds its own election every five years, on a calendar determined by the date of the previous election. National synchronisation is absent — unlike the Portuguese setting studied by [Veiga and Veiga \(2007\)](#) — which generates the variation in treatment timing that identification strategies in this paper exploit.

4 Data

Political data come from the Italian Ministry of the Interior’s *storico amministratori comuni* database (2010–2024). For each municipality-year, we record the election year, the mayor’s age, gender, and educational attainment. The indicator `last_term` equals 1 for mayors serving their second (final) term, constructed using the Italian electoral term-limit rules applied to the full administrative record (2010–2024). Municipalities under extraordinary administration (*scioglimento*) are excluded for the duration of the commissariamento episode. Electoral timing is expressed in event time relative to each municipality’s own election year: $t = 0$ is the election year, $t - 1$ and $t - 2$ are the one and two years before the election, and $t + 1$, $t + 2$ are the two years that follow.

Land use data come from the ISPRA *Carta Nazionale del Consumo di Suolo* (CNCS), a nationally consistent annual raster product produced by the Italian Institute for Environmental Protection and Research. This is the first paper to use CNCS data to study political business cycles. The source is administratively independent: ISPRA is a national environmental agency with no reporting relationship to municipal governments, and land-cover classifications are derived from remote sensing rather than municipal permit registries. This independence is methodologically valuable: if incumbents manipulate building permits as an electoral instrument, permit-based counts are endogenous to the very behaviour we wish to measure, whereas satellite-detected construction activity is not. The raster has a 10 m spatial resolution and is updated annually from 2015 onwards by comparing multi-temporal satellite imagery against the previous year’s classification. Each pixel is assigned to one of the categories in the ISPRA hierarchical taxonomy.¹

Among the many land-use categories tracked by ISPRA, construction sites (*cantieri*, code 122) are the most directly sensitive to short-run electoral manipulation. ISPRA records a plot as a construction site as soon as land is cleared and fenced — before any building is erected — so a permit decision translates into a satellite-detectable change within weeks. Other categories, such as buildings and paved roads, register only when construction is completed: a process spanning several years. Municipal aggregates are computed by clipping each annual raster layer to municipal boundaries and summing pixel areas in square metres.

Figure 1 illustrates the measurement approach and the core result in visual form. The left panel shows the full ISPRA pixel classification for Sezzadio (AL, Piemonte;

¹ISPRA distinguishes between *permanent* consumed land — completed hard infrastructure including buildings, paved roads, railways, airports, ports, parking lots, landfills, and extraction sites — and *reversible* consumed land — surfaces that could in principle revert to natural use, including construction sites, unpaved roads, photovoltaic panels, and greenhouses. Total consumed land is the sum of the two.

population 1,257), a small northern municipality that held a mayoral election in 2019. The right panel places the two years immediately flanking the election — 2018 ($t - 1$, year before) and 2020 ($t + 1$, year after) — side by side using the same colour scheme. The increase in active construction sites (orange) between the two panels is the within-municipality change that we estimate across the full sample.

The panel covers 2015–2024, yielding $7,761 \text{ municipalities} \times 10 \text{ years} = 77,610$ observations.

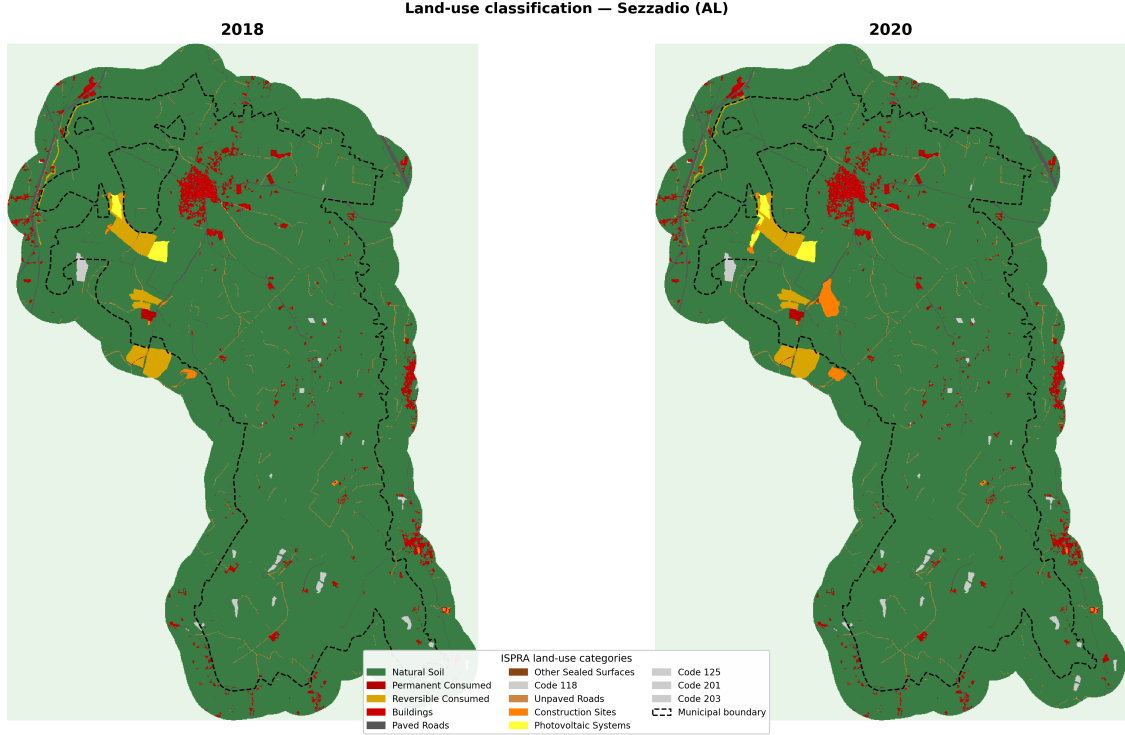


Figure 1: ISPRA land-use raster — Sezzadio (AL, Piemonte), population 1,257. Side-by-side comparison of 2018 ($t - 1$, year before the election) and 2020 ($t + 1$, year after). The increase in orange construction-site pixels between the two panels — from $25,600 \text{ m}^2$ to $176,300 \text{ m}^2$ — illustrates the post-election mandate cycle documented in Section 6. Municipal boundary shown as dashed line. Source: ISPRA *Carta Nazionale del Consumo di Suolo* (10 m resolution).

5 Identification Strategy

The workhorse specification in the PBC literature is a TWFE panel regression with election-year dummies, municipality fixed effects absorbing all time-invariant confounders, and year fixed effects absorbing common national shocks. [Veiga and Veiga \(2007\)](#) apply this to Portuguese municipalities; [Bracco et al. \(2024\)](#) and [Ferraresi et al. \(2024\)](#) apply the same framework to Italian road investment. Where elections are nationally synchronised and the treatment indicator switches on permanently, this approach identifies the pre-electoral build-up cleanly. Neither condition holds in

the Italian municipal panel.

Problem 1 - Staggered timing and forbidden comparisons. Italian municipal elections are staggered: each municipality votes every five years on a date determined by its own electoral history. Goodman-Bacon (2021) proves that TWFE in a staggered setting equals a weighted average of all pairwise 2×2 DiD comparisons: $\hat{\beta}^{\text{TWFE}} = \sum_k \sum_{j \neq k} \hat{w}_{kj} \hat{\beta}_{kj}^{2 \times 2}$ where $\hat{\beta}_{kj}^{2 \times 2}$ compares cohort k against cohort j in specific sub-windows, with implicit weights $\hat{w}_{kj}$ determined by cohort sizes and variance. Some of these comparisons are *forbidden*: when cohort k is compared against an earlier-treated cohort, the “control” units are already in a post-electoral phase and their outcomes reflect the effects of *their* election. Baker et al. (2022) show that when true treatment effects vary across cohorts — the empirically plausible case, since a 2016 election takes place in a different economic environment than a 2021 one — TWFE can inflate Type I error rates to 50–75% and can even reverse the sign of the true average effect.

Problem 2 - Recurring (pulse) treatment. The majority of modern DiD estimators designed to fix Problem 1 — Callaway and Sant’Anna (2021), Sun and Abraham (2021), Wooldridge (2021) — were built for *absorbing* treatment: once treated, always treated. Italian municipal elections violate this condition. Treatment equals 1 only in election years and *recurs* every five years:

$$D_{iy} = \mathbf{1}[y \in \{e_i, e_i + 5, e_i + 10, \dots\}] \quad (1)$$

where e_i is municipality i ’s first election year and y denotes the calendar year. This pulse structure has three practical consequences for identification. There is no never-treated comparison group: every Italian municipality holds an election every five years, so a unit used as “control” in calendar year y is itself treated at $y + 3$ (its next election) or was treated at $y - 2$ (its prior one). The “untreated” calendar years of municipality i include years that are $y + 1$ or $y + 2$ after its *previous* election; if the PBC generates post-electoral carryover, observationally untreated years are contaminated by the prior cycle. And TWFE forbidden comparisons (Problem 1) are doubly tainted: the “control” cohort may simultaneously be in its own post-electoral phase from an event-time perspective, making the Goodman–Bacon decomposition internally inconsistent. In our data this is visible: TWFE on the full panel for total consumption produces significant positive coefficients both pre- and post-election all significant at 1% (see first panel in Figure A1), the fingerprint of a long term trend that TWFE cannot separate from the electoral cycle.

5.1 Stacked DiD

To solve the two problems simultaneously, we implement a Stacked DiD design (Baker et al., 2022; Wing et al., 2024). The stacked DiD approach resolves both problems by explicitly constructing a clean comparison group for each election cohort. Specifically, for each cohort $g \in \{2016, 2017, 2018, 2019, 2020, 2021\}$, the treated group consists of all municipalities with election year $= g$, observed at event times $t \in \{-1, 0, +1, +2\}$ (where $t = y - g$ and y is the calendar year). The control group is municipalities with election year $= g + 3$. The $+3$ gap ensures two things simultaneously: the controls' next election ($g + 3$) falls after the event window $[g - 1, g + 2]$, so they are genuinely not-yet-treated within the window; and their prior election ($g + 3 - 5 = g - 2$) falls before the window start $g - 1$, so there is no carryover from the control's own previous cycle into the baseline period. This *prior-cycle contamination guard* is specific to the pulse-treatment setting — absorbing-treatment papers do not need it because their controls never cycle back into treatment.

To make this concrete: for cohort $g = 2017$, controls vote at 2020 and their prior election (2015) falls before the window start (2016), so at no point in $[2016, 2019]$ is the control municipality in an election year from its own cycle. For each cohort, we retain only calendar years $y \in [g - 1, g + 2]$: one pre-election year, the election year itself, and two post-election years. Extending to two pre-periods ($[g - 2, g + 2]$) would cause the control's prior election at $g - 2$ to coincide with the window start, re-introducing contamination.

Figures 2 and 3 illustrate the design. Figure 2 shows the cohort composition of the stacked sample: the 2019 cohort is the largest (3,782 treated municipalities) owing to the national synchronisation of elections that year; grey bars show years excluded because their event window extends outside the ISPRA data coverage (2015–2024). Figure 3 shows the event-window calendar for each cohort: the treated row (top) is colour-coded by event time; the controls row (middle) confirms that eligible controls' own election ($g + 3$) falls outside the window; the excluded row (hatched) shows municipalities disqualified from the control pool.

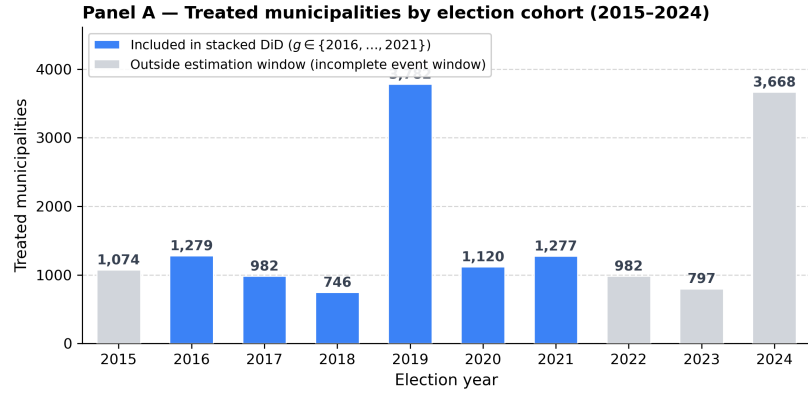


Figure 2: Number of municipalities with elections by year, 2015–2024. Blue bars: cohorts included as treated units in the stacked DiD ($g \in \{2016, \dots, 2021\}$). Grey bars: years excluded from the design because the event window $[g - 1, g + 2]$ extends outside the ISPRA data coverage (2015–2024). The 2024 spike mirrors 2019 owing to the five-year electoral cycle.

Stacked DiD Design — Cohort Structure and Clean Control Groups

Panel B — Event-window calendar: treated, eligible controls, and excluded municipalities

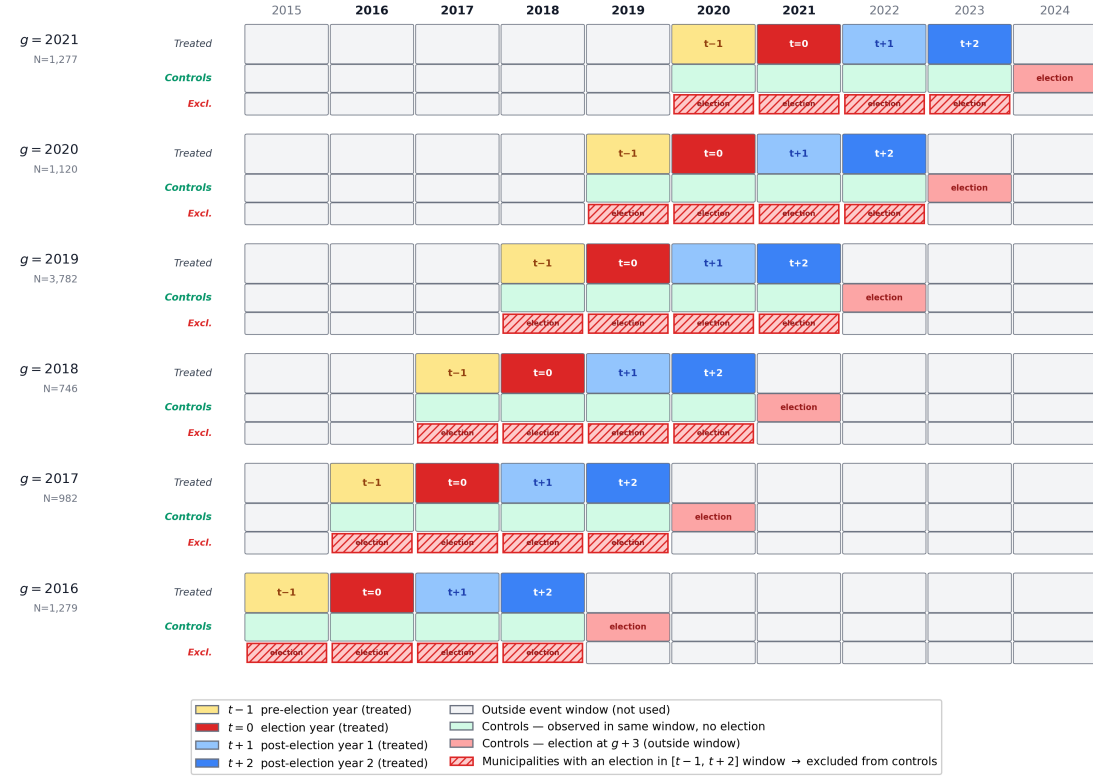


Figure 3: Event-window calendar for the stacked DiD design. Each block shows one election cohort g . *Treated* (top sub-row): colour-coded by event time ($t - 1$ yellow, $t = 0$ red, $t + 1$ light blue, $t + 2$ blue). *Controls* (middle sub-row): municipalities with election at $g + 3$, whose own election falls outside the estimation window. *Excluded* (bottom sub-row, hatched): municipalities with any election in $[g - 1, g + 2]$. Calendar years in **bold** are cohort election years.

The six sub-experiments are stacked into a single dataset ($\approx 70,000$ observations). A municipality appearing as a treated unit in one stack and as a control in another receives a separate baseline via $stack \times municipality$ fixed effects μ_i^s ; $stack \times year$ fixed effects ν_t^s absorb common shocks separately within each sub-experiment, so that a national event (eg. the Superbonus surge in 2020) is not double-counted across stacks. The pooled stacked regression is:

$$Y_{it}^s = \sum_{\tau \in \{-1, +1, +2\}} \delta_\tau D_i^s \cdot \mathbf{1}[t = \tau] + \mathbf{x}_{it}' \gamma + \mu_i^s + \nu_t^s + \varepsilon_{it}^s \quad (2)$$

where s indexes the stack (sub-experiment centred on cohort g), $t = y - g$ is event time relative to the cohort election year (with $t = 0$ denoting year g), τ indexes the event-time dummies, $D_i^s = 1$ for treated municipalities and 0 for controls, and the omitted baseline is $t = 0$. Standard errors are clustered at the municipality level.

The outcome variable is $Y_{it}^s = \ln(\text{construction sites}_{it} + 1)$.

Each $\hat{\delta}_\tau$ estimates a weighted average of cohort-specific average treatment effects on the treated (ATT) where the weights are proportional to cohort size; the 2019 cohort (3,782 treated municipalities) receives the largest weight. The identifying assumption is parallel trends within each stack: absent the election, treated and control municipalities in the same cohort would have followed the same trend in land use. Because controls share the same five-year electoral cycle and legal framework as the treated group, this assumption is more credible than in the full TWFE where controls can be at any point in their own cycle.² The pre-electoral coefficient $\hat{\delta}_{-1}$ serves as the empirical test.

6 Results

Figure 4 reports the main estimates from the stacked DiD estimator, implemented as described in Section 5³. For each election cohort g , treated municipalities are matched to a clean control group (municipalities holding their next election at $g + 3$, with no election within the event window), and the stacked sub-experiments are estimated jointly with interacted fixed effects that prevent cross-cohort contamination. The stacked dataset contains 76,132 observations.

The headline finding is a *post-election mandate cycle* in construction activity: construction is suppressed in the year before the election and surges significantly in the two years that follow. The subsections below establish the mechanism driving this pattern, quantify its amplitude, and characterise which municipalities it concentrates in.

²Borusyak et al. (2024) propose an efficient imputation estimator for event-study designs that would in principle yield lower-variance ATT estimates under the same parallel-trends assumption. Their approach requires a clean pool of never-treated or not-yet-treated controls; in our setting, that condition holds only by virtue of the prior-cycle contamination guard imposed by the stacking design, making the stacked DiD the natural implementation vehicle.

³In the Appendix we report TWFE event-study estimates, both with and without controls (Section A). The TWFE results are inconsistent with the parallel trends assumption: significant positive coefficients appear both before and after the election, suggesting a long-term upward trend in land consumption that TWFE cannot separate from the electoral cycle. This motivates the Stacked DiD design, which explicitly constructs clean control groups for each cohort and imposes a prior-cycle contamination guard specific to recurring electoral cycles.

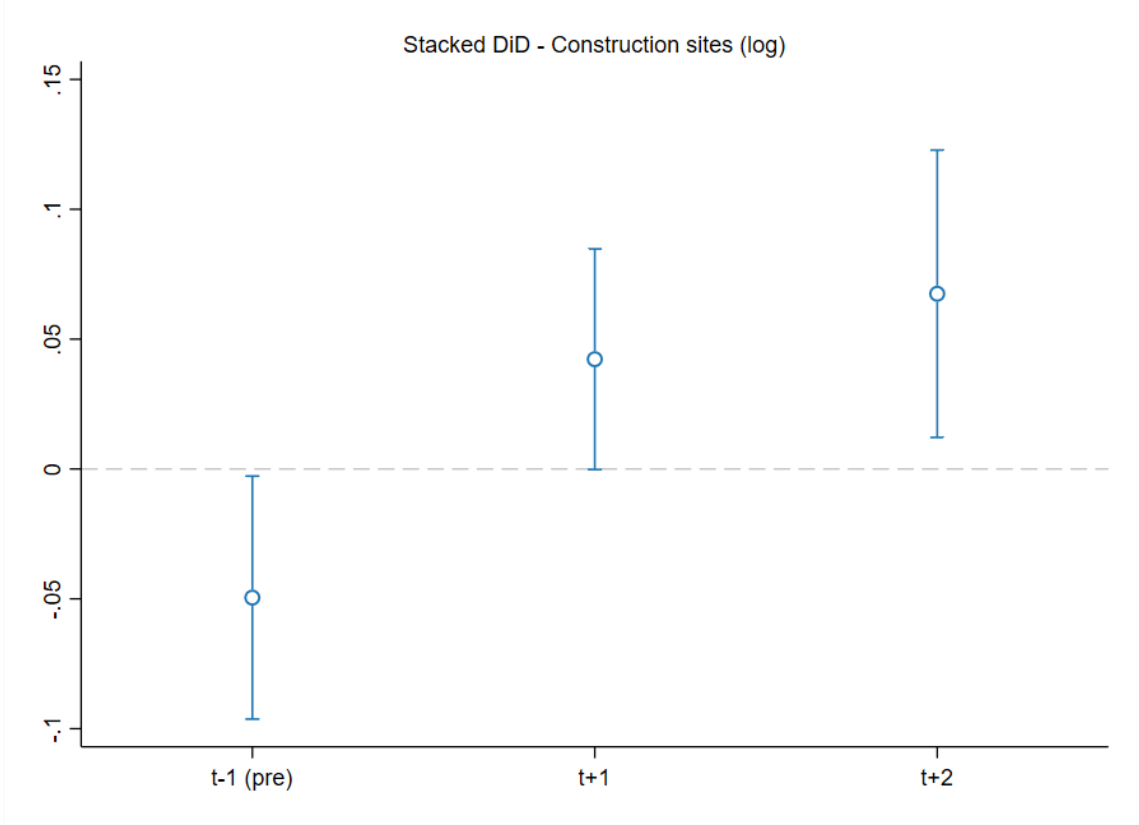


Figure 4: Stacked DiD event-study estimates — conditional specification for construction sites. Dots are point estimates; capped bars are 95% confidence intervals (SE clustered at municipality level). Omitted baseline: election year ($t = 0$). FE: $\text{stack} \times \text{municipality}$ and $\text{stack} \times \text{year}$.

Three results stand out. First, construction sites exhibit a statistically significant post-electoral expansion: $D_{+2} = 0.068^{**}$, corresponding to an approximately 7% increase in construction-site activity two years after the election relative to the election year itself. The year-one post-electoral effect is positive and marginally significant ($D_{+1} \approx 0.041^*$). This graduated build-up, stronger at $t+2$ than $t+1$, is consistent with the administrative lag between permit authorisation and physically observable site activity, and with the time needed for a newly or re-elected administration to translate its mandate into concrete planning decisions. Second, the pre-electoral coefficient D_{-1} is negative. We interpret this as an *electoral lull*: incumbents facing re-election deliberately slow disruptive construction activity — noise, road closures, scaffolding — in the year immediately before polling, accumulating a backlog of construction authorisations.⁴ As a sensitivity check, we apply the same specification to the

⁴This is apparently at odds with Bracco et al. (2024) and Ferraresi et al. (2024), who find road investment rising sharply at $t - 1$ in the same Italian setting. The reconciliation lies in what each measure captures. Those papers measure fiscal procurement and expenditure, which includes road *maintenance* on existing infrastructure: resurfacing, pothole filling, asphalt patching. ISPRA records only changes in land-cover status; maintenance on already-sealed road surfaces produces no new sealed pixels and is therefore invisible to our outcome variable. An incumbent can simultaneously increase maintenance spending on existing roads — generating the pre-electoral boom in fiscal

remaining ISPPRA categories — total consumed land, buildings, paved roads, other sealed surfaces, and permanent consumption — which are longer-run stock measures that only register at project completion. None shows a statistically significant electoral cycle. Full results are in Appendix Table A2 and Figure A3. These are predicted nulls, not incidental findings — they reflect the multi-year administrative timelines of buildings and roads, which render these categories unsuitable as within-term electoral instruments.

Interpreting the pre-electoral lull. The negative pre-electoral coefficient D_{-1} demands careful interpretation because it occupies a peculiar position in the stacked DiD design: it is simultaneously the event-study estimate for $t - 1$ and the *only* pre-period observation available to the estimator. Three alternative readings must therefore be confronted.

1. The first is a *parallel trends violation*: if construction in treated municipalities was already declining relative to controls before the election window, D_{-1} would capture a spurious pre-existing trend. Aware the PTA is untestable, we tried to test the assumption using the TWFE estimator - which, unlike the Stacked DiD - can accommodate a two-period pre-window without depleting the control group. Figure A2 in Appendix displays event-study plots for construction sites in the baseline and conditional TWFE with both $t - 2$ and $t - 1$ included as free pre-treatment coefficients. The $t - 2$ coefficient is $\hat{\delta}_{-2} = -0.006$ (s.e. = 0.015) in the baseline and $\hat{\delta}_{-2} = -0.004$ (s.e. = 0.015) in the conditional specification, both far from statistical significance ($p > 0.5$ in both cases). In contrast, if the negative D_{-1} reflected a systematic divergent trend, we would expect $t - 2$ to also be negative and significant. The null at $t - 2$ establishes that the data exhibit a flat pre-trend at least two periods before the election, supporting the parallel trends assumption and validating the interpretation of D_{-1} as a structural, one-period permit freeze rather than a trend artefact.
2. The second interpretation is an *anticipation effect*: forward-looking mayors might begin slowing down construction even at $t - 2$ or earlier, in anticipation of the approaching election. This would predict a symmetric or stronger lull at $t - 2$, which the data again contradict: a genuine declining trend would have produced a negative and significant $t - 2$; it does not and the lull is sharp and concentrated in the single pre-election year.

road investment — while deferring the opening of new construction sites on previously unsealed land, which is what our pre-electoral lull reflects. The two results measure different activities and are complementary rather than contradictory. This reading is further supported by [Kapur and Vaishnav \(2011\)](#), who document that construction activity in Indian states contracts in election months.

3. The third and preferred reading is a genuine *electoral-lull effect* that is an integral part of the mandate cycle itself. The same re-election incentive that produces the post-electoral surge also produces its mirror image in the preceding year: an incumbent who wishes to claim credit for infrastructure improvements after re-election has every reason to suppress visible construction disruption in the run-up to the vote. We test this mechanism directly in the following section: if the re-election incentive is the operative channel, removing it — as happens when a mayor faces a binding term limit — should reverse or eliminate the lull.

6.1 Mechanism: The Re-election Incentive

Italian electoral law imposes a two-consecutive-term limit on mayors. A mayor in their first term faces a binding re-election incentive; under the mandate theory, they should exhibit the full cycle: an electoral lull at $t - 1$ followed by a post-electoral surge. A second-term mayor cannot stand again and faces no re-election constraint; without the mandate incentive, the same cycle should not appear, or should reverse.

Figure 5 presents conditional Stacked DiD estimates split by a dummy variable indicating whether the mayor is in their last term. The results are strikingly asymmetric and constitute the sharpest identification evidence in this paper. Among re-election-eligible mayors, the mandate cycle is clean and significant: $D_{-1} = -0.060^*$ (s.e. = 0.032), $D_{+1} = 0.049^*$ (s.e. = 0.027), $D_{+2} = 0.084^{**}$ (s.e. = 0.035). The electoral lull and the post-mandate surge appear exactly as predicted. Among lame-duck mayors, the pattern reverses sign: $D_{-1} = +0.202^*$ (s.e. = 0.111), with D_{+1} and D_{+2} statistically null. Outgoing mayors — freed from the re-election constraint — accelerate construction before the election rather than suppressing it. This is consistent with a political-debt mechanism: incumbents who cannot stand again may rush to complete or reward projects for aligned contractors before leaving office — a pattern related to the procurement evidence in [Coviello and Gagliarducci \(2017\)](#), who document that longer mayoral tenure raises the probability of repeated contractor selection by 25 percentage points. The sign reversal is sharp and theoretically motivated: the same electoral proximity that induces a lull when re-election is at stake produces a boom when it is not. This rules out any interpretation of the full-sample D_{-1} as a mechanical or seasonal artifact of construction-industry dynamics: if it were, the sign would not flip at the term-limit boundary.

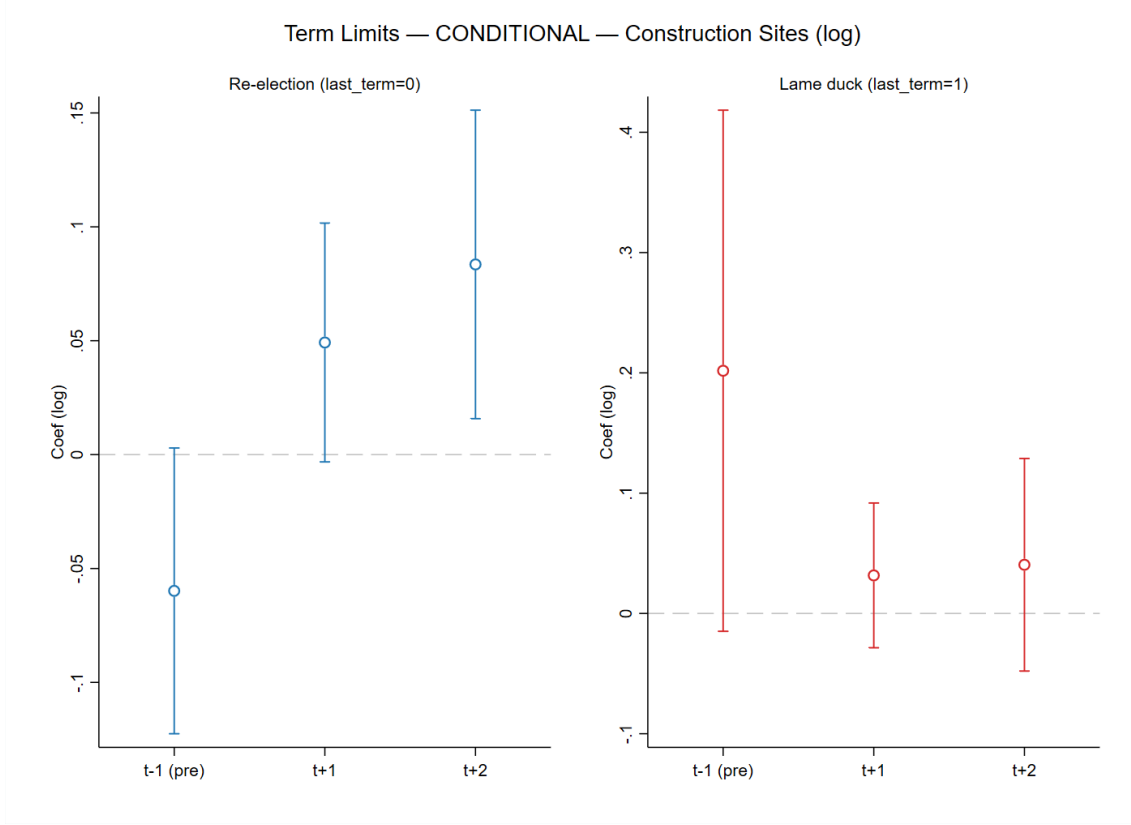


Figure 5: Stacked DiD — mechanism test by term limit, conditional specification, `log_constr`. Left panel: re-election-eligible mayors (`last_term = 0`, $N = 57,331$). Right panel: lame-duck mayors (`last_term = 1`, $N = 15,240$). The full mandate cycle appears among first-term mayors; lame-duck mayors show a large positive pre-electoral coefficient consistent with a political-debt mechanism. The sign reversal across the term-limit boundary confirms that the re-election incentive drives the cycle.

Amplitude of the Mandate Cycle The estimates in Figure 4 measure construction activity relative to the election year. To quantify the full swing of the mandate cycle, from the pre-electoral lull to the post-electoral peak, we re-estimate the conditional Stacked DiD using $t - 1$ as the omitted reference period. This normalisation directly answers the economic question: by how much does post-election construction exceed the trough recorded in the year before the election? In the Appendix Figure A4 presents the results. The election-year coefficient D_0 is 0.050^{**} (s.e. = 0.024), indicating that construction activity begins recovering at the election itself. The post-electoral surge is large and precisely estimated: $D_{+1} = 0.092^{***}$ (s.e. = 0.031) and $D_{+2} = 0.117^{***}$ (s.e. = 0.033), corresponding to a 10% and 12% increase in construction sites above the pre-electoral trough in years one and two after the election, respectively. This amplitude is economically substantial, and consistent with the re-election mechanism established before: the backlog of deferred authorisations is deployed once the mandate is renewed, and the surge accumulates

over two years as projects approved in the weeks following the election become visible construction sites only in the months and years that follow.

6.2 Heterogeneity

Municipality Size Municipal governments in Italy face different institutional constraints depending on population size. In smaller municipalities, the mayor directly controls the planning office and can influence permit decisions on short timescales; in larger municipalities, bureaucratic layers attenuate this direct control.

Figure 6 reports Stacked DiD estimates (baseline specification) for `log_constr` across four population bins. The electoral cycle is entirely concentrated in the smallest bin ($\leq 3,000$ inhabitants, $N = 4,648$ treated municipalities): $D_{-1} = -0.037$, $D_{+1} = 0.071^*$ (s.e. = 0.037), $D_{+2} = 0.117^{**}$ (s.e. = 0.049). For all larger bins (3,001–5,000; 5,001–15,000; $> 15,000$ inhabitants), every coefficient is statistically null.

This finding is economically interpretable: in Italian municipalities with fewer than 3,000 inhabitants, the mayor exercises direct day-to-day authority over the technical office responsible for permit issuance. The absence of a cycle in larger municipalities suggests that the electoral manipulation of construction activity is a small-municipality phenomenon driven by personalised mayoral control over the permitting process, not a system-wide feature of Italian land governance.

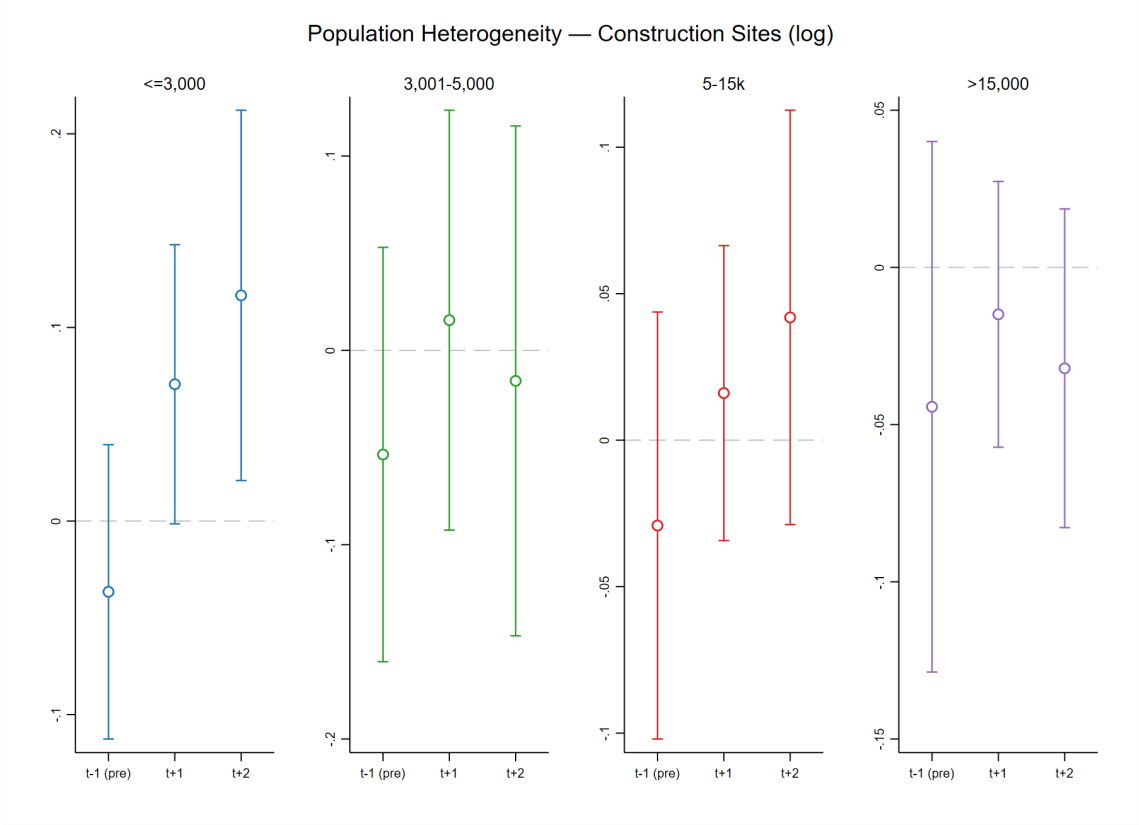


Figure 6: Stacked DiD heterogeneity by municipality size — baseline specification, `log_constr`. Four population bins: $\leq 3,000$; 3,001–5,000; 5,001–15,000; $> 15,000$ inhabitants. The electoral PBC is statistically significant only in the smallest bin, where mayors retain direct administrative control over permit issuance.

Geographic Heterogeneity Figure 7 splits the baseline Stacked DiD by macro-region (North Italy: Val d’Aosta, Piemonte, Liguria, Lombardia, Emilia-Romagna, Veneto, Trentino Alto-Adige, Friuli-Venezia Giulia; Centre-South: remaining regions). The post-electoral surge is driven by northern municipalities: $D_{+2} = 0.088^{**}$ (s.e. = 0.042) in the North, compared to a statistically null $D_{+2} = 0.029$ (s.e. = 0.040) in the Centre-South. The pre-electoral coefficient is negative in both sub-samples but not significant in either.

The geographic concentration in the North is consistent with two interpretations. First, northern municipalities have on average higher construction activity and a more active private construction sector, making permit decisions more consequential as an electoral instrument. Second, informality and political clientelism operate differently in the Centre-South, where electoral manipulation may take forms (e.g., targeted transfers, employment in public agencies) that are not captured by land-use data. We do not attempt to distinguish these mechanisms here.

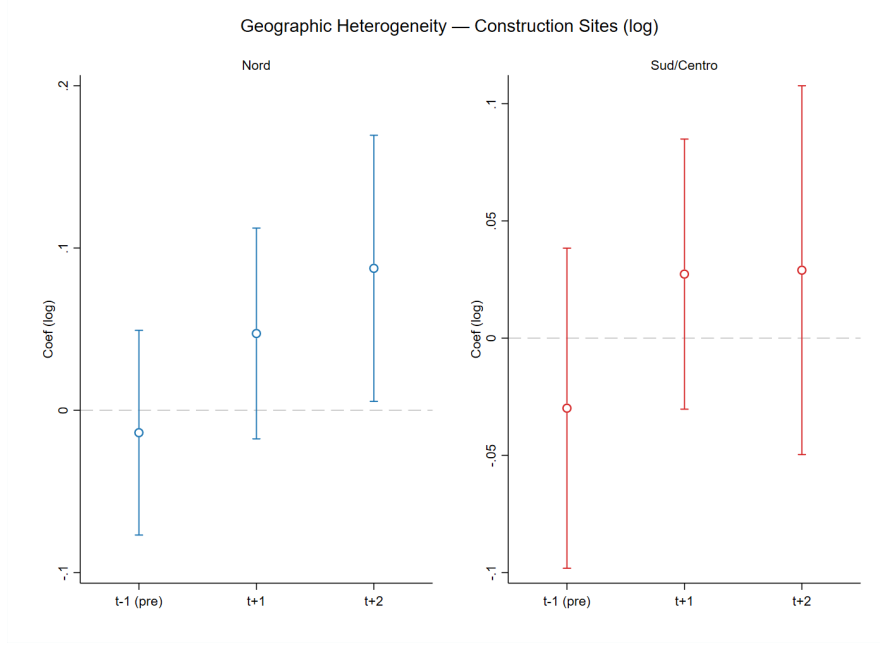


Figure 7: Stacked DiD heterogeneity by macro-region — baseline specification, `log_constr`. Left panel: North Italy (ISTAT regions 1–8). Right panel: Centre-South. The post-electoral surge is statistically significant only in northern municipalities.

6.3 Robustness

Cohort Exclusion The 2019 election cohort occupies an unusual position in the estimation sample: its post-electoral window ($t + 1 = 2020$, $t + 2 = 2021$) falls entirely within the COVID-19 disruption period and the subsequent construction-sector boom. The *Superbonus 110%* programme (*Decreto Rilancio*, DL 34/2020) is a prominent feature of those years, but it is worth stressing that the Superbonus was a tax credit claimed directly by private citizens and condominiums for renovation of existing buildings; it was not administered through municipal planning offices and provides no channel through which mayors could directly influence our outcome variable. The concern is rather about the extraordinary macroeconomic conditions of 2020–2021 more broadly — COVID-19 lockdowns and the ensuing recovery, an unprecedented surge in construction-sector demand, and supply-chain disruptions — which could affect construction activity in ways that are imperfectly controlled by year fixed effects. Stack $\times$ year fixed effects absorb shocks that are common to all municipalities within a cohort in a given year; they cannot rule out that municipality-specific responses to an unusual economic environment varied systematically between treated municipalities (entering their post-electoral phase in 2020–2021) and their controls (still two years before their own 2022 election). We therefore verify that the main results survive when the most-exposed cohort is excluded. The 2018 cohort is partially exposed ($t + 2 = 2020$) and is excluded in the most conservative specification.

Figure 8 presents the three event-study plots for the full sample and the two restricted samples. Two patterns emerge. First, D_{-1} *strengthens* when the 2019 cohort is excluded (-0.071^{***} , versus -0.050^{**} in the full sample) and remains significant when both 2018 and 2019 are excluded (-0.068^{**}), confirming that the pre-electoral lull is not an artefact of conditions specific to those years. Second, the post-electoral coefficients D_{+1} and D_{+2} remain positive across all samples and are of comparable economic magnitude ($D_{+2} = 0.048$ – 0.053 in the restricted samples, versus 0.068 in the full sample). Confidence intervals widen because the 2019 cohort is by far the largest in the estimation sample (3,782 treated municipalities), and its exclusion removes approximately 24% of observations, mechanically reducing precision. The loss of conventional significance reflects reduced power rather than instability of the underlying effect. The electoral cycle in construction sites is not an artefact of the unusual macroeconomic conditions of 2020–2021.

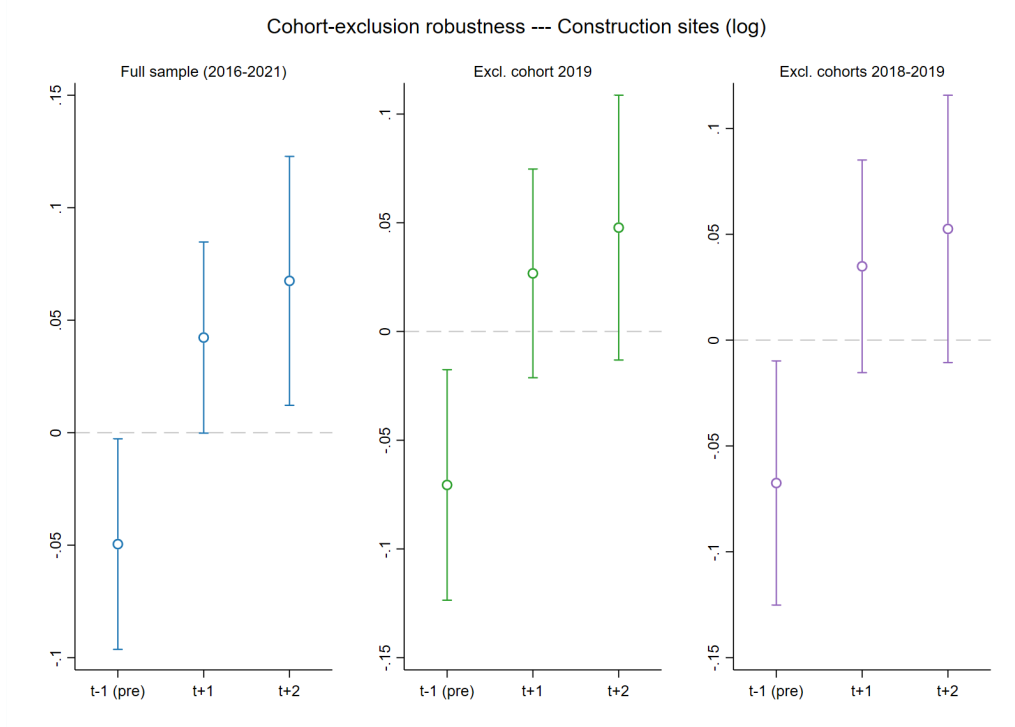


Figure 8: Cohort-exclusion robustness — conditional Stacked DiD for construction sites. Left: full sample (cohorts 2016–2021). Centre: excluding the 2019 cohort, whose post-electoral window ($t+1 = 2020$, $t+2 = 2021$) coincides with the COVID-19 disruption period and the subsequent construction-sector boom. Right: excluding both 2018 and 2019.

7 Conclusion

This paper makes two contributions to the empirical PBC literature. On the data side, we introduce ISPRA satellite-derived land-cover data as an outcome variable — a

source that is administratively independent and immune to retroactive manipulation by the governments under study. On the identification side, we implement a Stacked DiD design with a prior-cycle contamination guard specific to recurring electoral treatments, addressing a problem that has been overlooked in PBC empirical work.

The main finding is a *post-election mandate cycle* in construction activity. Italian first-term mayors suppress construction in the year before an election — a deliberate electoral lull — and accelerate it in the two years that follow; re-normalised to the pre-electoral trough, the cycle amplitude is approximately 12 percentage points. This pattern is structurally tied to the re-election incentive: lame-duck mayors, who cannot stand again, exhibit the opposite pattern, rushing construction before the election consistent with a political-debt mechanism toward aligned contractors. The sign reversal at the term-limit boundary is the central identification result of the paper: it rules out any mechanical or seasonal interpretation of the pre-electoral lull.

The finding is specific to construction sites — the only land-use category whose administrative timeline is short enough to be deployed as a within-term electoral instrument. Buildings, roads, and other permanently sealed surfaces only appear in satellite data at project completion and show no electoral cycle. These are predicted nulls, not incidental ones.

The cycle is entirely concentrated in municipalities with fewer than 3,000 inhabitants, where the mayor exercises direct day-to-day authority over permit issuance, and in northern Italy. A direct policy implication follows: simplifying or delegating permit authority in small municipalities would attenuate this channel of electoral manipulation without requiring reform of the electoral system itself.

These results have two broader implications. First, when fiscal manipulation space is compressed — as it has been in Italy since the 2008 abolition of the primary-residence property tax — elected officials redirect electoral activity toward regulatory channels such as land-use authorisation. Electoral manipulation of local public goods is not limited to fiscal instruments. Second, satellite-derived land-cover data offer a path to studying electoral cycles in physical outcomes across countries and time periods where fiscal data is unreliable or administratively endogenous.

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A Baseline TWFE Event Study

As a transparent benchmark, we show the workhorse two-way fixed effects (TWFE) event-study estimator applied to the full panel. Figure A1 presents the conditional specification (municipality and year fixed effects, plus the full set of controls) for all six land-use outcomes.⁵ The pattern is unambiguous: across total consumed land, buildings, paved roads, other sealed surfaces, and permanent consumption, the electoral-cycle coefficients are statistically indistinguishable from zero throughout the window. The sole exception is construction sites. For construction sites in the conditional TWFE, the pre-electoral coefficient at $t - 2$ is $\hat{\beta}_{-2} = -0.004$ (s.e. = 0.015), not significantly different from zero. The coefficient at $t - 1$ is similarly null ($\hat{\beta}_{-1} = -0.024$, s.e. = 0.015). Post-election activity shows a positive and statistically significant surge at $t + 2$ ($\hat{\beta}_{+2} = 0.028^{**}$, s.e. = 0.014), with a marginally positive but insignificant estimate at $t + 1$ ($\hat{\beta}_{+1} = 0.014$, s.e. = 0.014).

Two features of the TWFE results deserve emphasis. First, several other outcomes (total consumed land, roads, permanent consumption) yield statistically significant coefficients at conventional levels, but their point estimates range from 0.0004 to 0.0021 — effects of less than 0.2 percentage points with no plausible economic interpretation. As shown in the stacked DiD section, these effects disappear entirely under stacked DiD, suggesting they reflect the negative-weight bias documented by [Goodman-Bacon \(2021\)](#) rather than genuine electoral manipulation. Second, the null at $t - 2$ is a key diagnostic result: we return to it in the next subsection as the primary evidence for parallel trends.

⁵Full coefficient tables including all control variables are reported in Appendix Table A1.

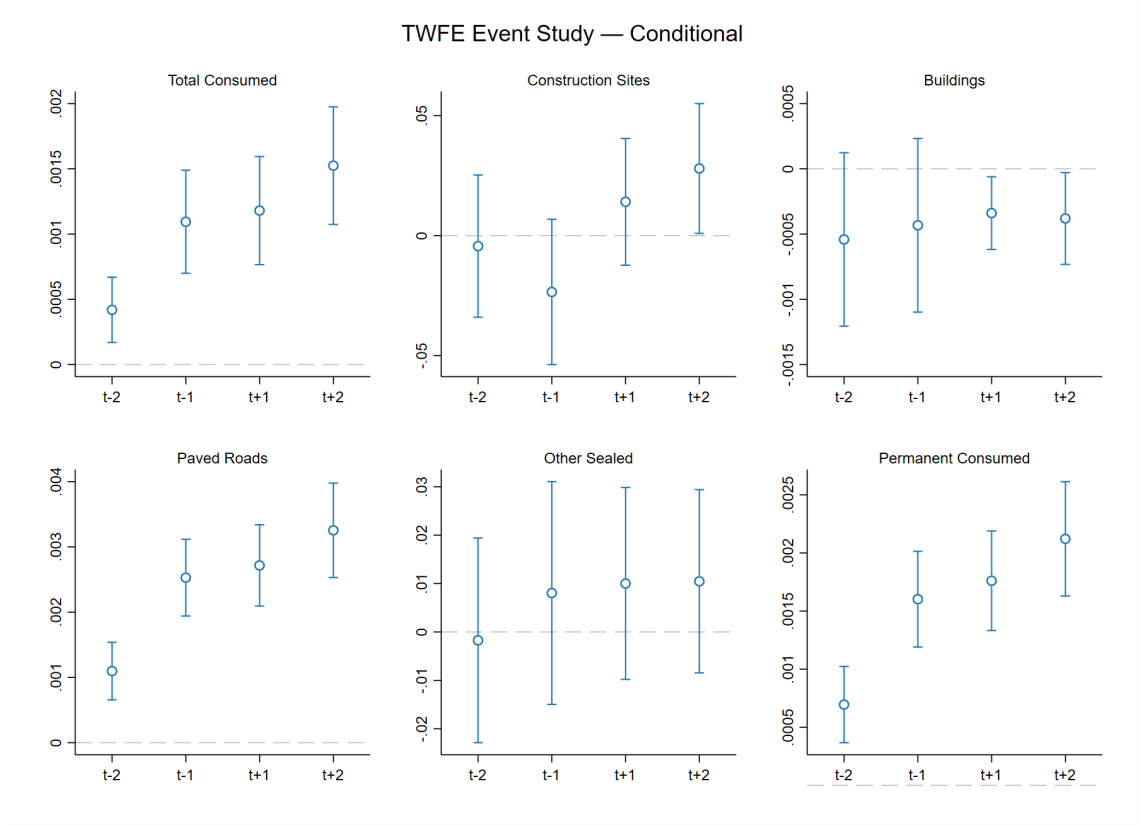


Figure A1: TWFE event-study estimates — conditional specification. Each panel shows one land-use outcome. Dots are OLS point estimates; capped bars are 95% confidence intervals clustered at the municipality level. The omitted baseline is the election year ($t = 0$). Municipal and year fixed effects absorbed. Controls: log population, gender, age at election, education, electoral system (majoritarian dummy), party bloc (left, right). Municipalities under extraordinary administration (*scioglimento*) excluded. $N = 75,816$.

B Full Regression Tables

This appendix reports full coefficient tables for the two main specifications discussed in the text. Panel A shows the election-cycle coefficients (the causal estimates of interest); Panel B shows the control-variable coefficients included in the conditional specifications. Standard errors are clustered at the municipality level throughout.

Table A1: Two-Way Fixed Effects — Full Coefficients (Conditional Specification)

	(1) Consumed	(2) Constr	(3) Buildings	(4) Roads	(5) Sealed	(6) Perm
<i>Panel A: Election-cycle coefficients (baseline: $t = 0$)</i>						
$t - 2$	0.0004*** (0.0001)	-0.0044 (0.0151)	-0.0005 (0.0003)	0.0011*** (0.0002)	-0.0017 (0.0108)	0.0007*** (0.0002)
$t - 1$	0.0011*** (0.0002)	-0.0235 (0.0154)	-0.0004 (0.0003)	0.0025*** (0.0003)	0.0080 (0.0117)	0.0016*** (0.0002)
$t + 1$	0.0012*** (0.0002)	0.0141 (0.0135)	-0.0003** (0.0001)	0.0027*** (0.0003)	0.0100 (0.0101)	0.0018*** (0.0002)
$t + 2$	0.0015*** (0.0002)	0.0280** (0.0138)	-0.0004** (0.0002)	0.0033*** (0.0004)	0.0105 (0.0097)	0.0021*** (0.0003)
<i>Panel B: Control variables</i>						
ln Population	0.0202*** (0.0053)	-3.3672*** (0.4140)	0.0702*** (0.0055)	-0.0171*** (0.0057)	-1.2873*** (0.2922)	0.0377*** (0.0046)
Female mayor	-0.0001 (0.0005)	-0.0372 (0.0452)	-0.0006 (0.0004)	0.0000 (0.0005)	-0.0325 (0.0341)	-0.0004 (0.0005)
Mayor age	-0.0000 (0.0000)	-0.0010 (0.0016)	-0.0001 (0.0001)	0.0000 (0.0000)	0.0011 (0.0011)	0.0000 (0.0000)
University degree	0.0001 (0.0003)	-0.0542 (0.0339)	-0.0002 (0.0003)	0.0005 (0.0004)	-0.0128 (0.0253)	0.0002 (0.0003)
Majoritarian system	0.0016 (0.0016)	-0.0708 (0.0797)	-0.0001 (0.0007)	0.0015 (0.0027)	-0.3103** (0.1408)	0.0021 (0.0017)
Left-wing party	0.0004 (0.0007)	0.0868 (0.0557)	0.0013*** (0.0004)	-0.0010 (0.0006)	0.0575 (0.0478)	0.0002 (0.0006)
Right-wing party	0.0012* (0.0007)	-0.1028** (0.0500)	0.0002 (0.0007)	0.0009 (0.0010)	-0.0694 (0.0440)	0.0010 (0.0009)
N	75,816	75,816	75,816	75,816	75,816	75,816

Notes: Two-way fixed effects (TWFE) estimator (**reghdfe**). Dependent variables are log-transformed land-use outcomes: total consumed land, construction sites, buildings, paved roads, sealed surfaces, and permanent consumed land. Fixed effects: municipality + year. Standard errors clustered at municipality level. Election year ($t = 0$) is the omitted baseline; $t - 2$, $t - 1$ are pre-election years; $t + 1$, $t + 2$ are post-election years. Municipalities under extraordinary administration (*scioglimento*) are excluded from the sample. Left-wing and right-wing party dummies are relative to civic/independent lists. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table A2: Stacked Difference-in-Differences — Full Coefficients (Conditional Specification)

	(1) Consumed	(2) Constr	(3) Buildings	(4) Roads	(5) Sealed	(6) Perm
<i>Panel A: Election-cycle coefficients (baseline: $t = 0$)</i>						
D_{-1}	0.0000 (0.0001)	-0.0495** (0.0239)	-0.0002 (0.0001)	0.0000 (0.0001)	-0.0023 (0.0156)	-0.0000 (0.0001)
D_{+1}	0.0001 (0.0002)	0.0423* (0.0217)	0.0000 (0.0001)	0.0000 (0.0001)	0.0026 (0.0134)	0.0001 (0.0001)
D_{+2}	0.0001 (0.0002)	0.0675** (0.0282)	-0.0001 (0.0001)	0.0001 (0.0001)	-0.0058 (0.0186)	0.0002* (0.0001)
<i>Panel B: Control variables</i>						
ln Population	0.0189*** (0.0030)	-1.3581*** (0.4050)	0.0365*** (0.0030)	0.0094*** (0.0018)	-0.6726*** (0.2525)	0.0325*** (0.0024)
Female mayor	0.0004 (0.0005)	-0.0527 (0.0452)	-0.0002 (0.0003)	0.0002 (0.0002)	-0.0380 (0.0240)	-0.0001 (0.0003)
Mayor age	-0.0000 (0.0000)	-0.0023 (0.0016)	0.0000 (0.0000)	-0.0000 (0.0000)	0.0000 (0.0009)	0.0000 (0.0000)
University degree	-0.0002 (0.0002)	-0.0630* (0.0347)	-0.0000 (0.0002)	0.0001 (0.0001)	0.0113 (0.0204)	0.0001 (0.0002)
Majoritarian system	0.0001 (0.0008)	-0.0871 (0.1220)	0.0008** (0.0004)	-0.0002 (0.0005)	-0.1857 (0.1904)	-0.0000 (0.0005)
Left-wing party	0.0005 (0.0005)	0.0644 (0.0579)	0.0010*** (0.0003)	0.0003 (0.0002)	0.0144 (0.0490)	0.0008** (0.0003)
Right-wing party	0.0007 (0.0004)	-0.0431 (0.0581)	0.0008** (0.0004)	0.0005 (0.0004)	-0.0437 (0.0402)	0.0008* (0.0004)
N	75,908	75,908	75,908	75,908	75,908	75,908

Notes: Stacked difference-in-differences estimator following Baker et al. (2022). Dependent variables are log-transformed land-use outcomes: total consumed land, construction sites, buildings, paved roads, sealed surfaces, and permanent consumed land. Fixed effects: stack $\times$ municipality + stack $\times$ year. Standard errors clustered at municipality level. Election year ($t = 0$) is the omitted baseline; D_{-1} = one year before the election; D_{+1} , D_{+2} = one and two years after. Clean-control design: control units voted at $g + 3$ (exactly). Cohorts 2016–2021. Left-wing and right-wing party dummies are relative to civic/independent lists. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

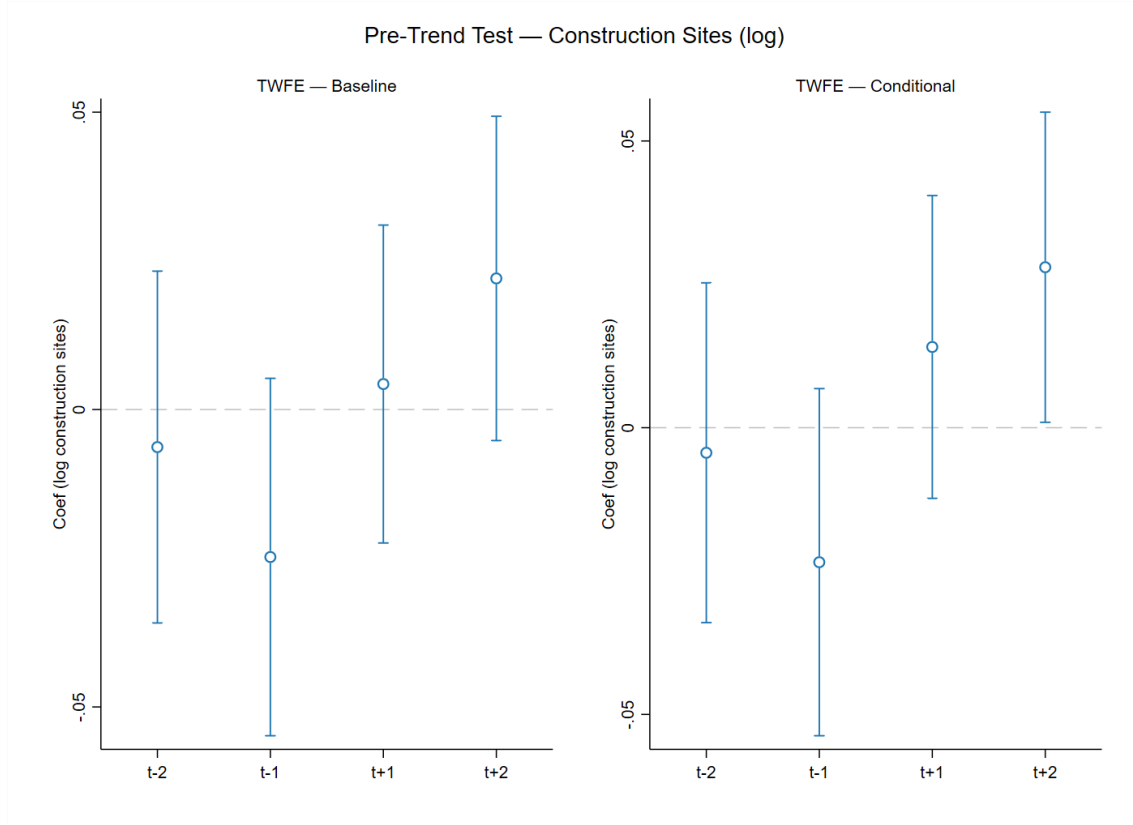


Figure A2: Parallel trends test — TWFE event study for `log_constr` with extended pre-window ($t-2$, $t-1$). Left panel: baseline (no controls). Right panel: conditional. The $t-2$ coefficient is statistically indistinguishable from zero in both specifications, supporting the parallel trends assumption. $N = 76,156$ (baseline), 75,816 (conditional).

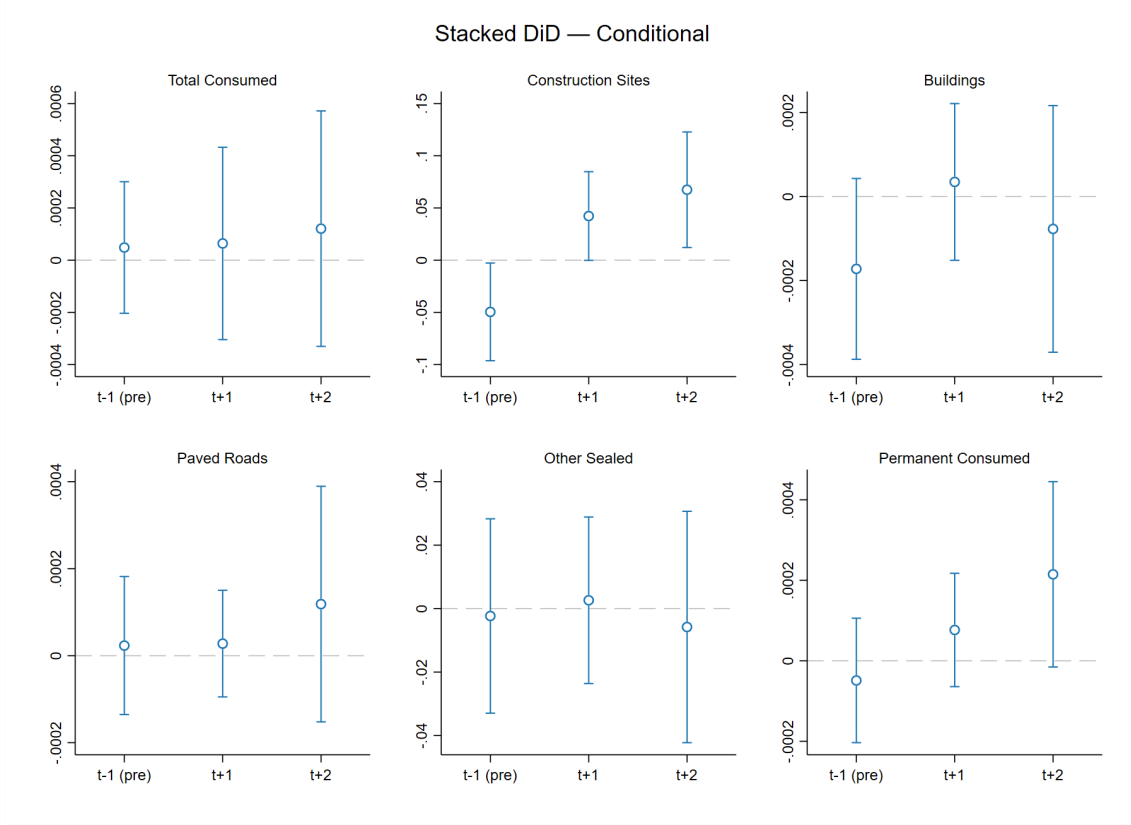


Figure A3: Sensitivity check: Stacked DiD conditional specification applied to all ISPra land-use categories. Construction sites (column 2) is the primary outcome; remaining columns are longer-run stock measures (total consumption, buildings, paved roads, sealed surfaces, permanent consumption). Only construction sites displays a significant electoral cycle, consistent with the prediction that short-horizon permit activity responds to electoral incentives while multi-year completion stocks do not. Dots: point estimates; capped bars: 95% CI (SE clustered at municipality level). Omitted baseline: $t = 0$. FE: $\text{stack} \times \text{municipality}$ and $\text{stack} \times \text{year}$.

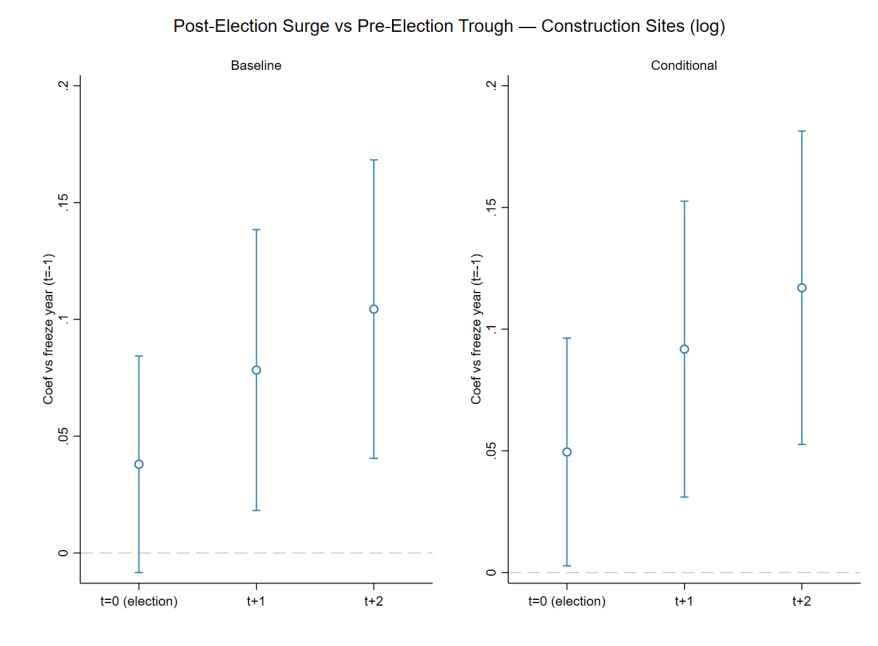


Figure A4: Stacked DiD — amplitude specification with $t - 1$ as baseline. Outcome: `log_constr`. Left panel: baseline (no controls). Right panel: conditional. Coefficients measure the surge in construction sites relative to the pre-electoral freeze year. Capped bars: 95% CI. $N = 76,132$ (baseline), $75,908$ (conditional).